ADVANCED LINEAR ALGEBRA

ASSIGNMENT - 4

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1. Define Vector Space.
2. Define Subspace.
3. Is R^n a vector space over the field of real numbers R?
4. Which of the following sets are subspaces of the vector space R^3 over R? Justify your answer.
   (i) $W_1 = \{(a, b, c) \in R^3 \mid pa + qb + rc = 0,$ where p, q, r are fixed real numbers$\}$
   (ii) $W_2 = \{(\alpha, \beta, \gamma) \in R^3 \mid \alpha^2 + \beta^2 + \gamma^2 \leq 1\}$
   (iii) $W_3 = \{(x, y, z) \in R^3 \mid z = 2y, y = x + z\}$
5. Define linearly dependent (L.D.) and linearly independent (L.I.) set.
6. Test the sets for L.D. or L.I. in the linear space R^3 over R.
   (i) $S = \{(-1, 1, 2), (2, -3, 1), (10, -1, 0)\}$
   (ii) $T = \{(1, 3, 2), (1, -7, -8), (2, 1, -1)\}$
7. Define basis and dimension.
8. Show that $B = \{(1, 1, 1), (1, 0, 1), (1, -1, -1)\}$ is a basis of the vector space R^3 over R.
9. State Rank-Nullity Theorem.
10. Find all eigenvalues and basis of each eigenspace of the matrix:

$$A = \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix}$$

    Is A diagonalizable? If so, find matrix P such that $P^{-1}AP$ is a diagonal matrix.
11. Define inner product of any two vectors. What is the length of a vector? Define norm of a vector.
12. State Gram-Schmidt orthogonalization process.
13. Apply Gram-Schmidt process to vectors $\mathbf{v}_1 = (3, 0, 4)$, $\mathbf{v}_2 = (-1, 0, 7)$, $\mathbf{v}_3 = (2, 9, 11)$ to obtain an orthonormal basis.
14. Find an orthonormal basis of the subspace of R^4 spanned by vectors:

$$\mathbf{v}_1 = (1, 1, 1, 1), \quad \mathbf{v}_2 = (1, 2, 4, 5), \quad \mathbf{v}_3 = (1, -3, -4, -2)$$

15. Solve the following system using LU decomposition method:

$$\begin{aligned} 2x - 3y + 10z &= 3 \\ -x + 4y + 2z &= 20 \\ 5x + 2y + z &= -12 \end{aligned}$$

16. Solve the following system using LU decomposition method:

$$\begin{aligned} 5x - 2y + z &= 4 \\ 7x + y - 5z &= 8 \\ 3x + 7y + 4z &= 10 \end{aligned}$$

17. Define idempotent matrix, projection matrix, partition matrix, orthogonal matrix. State any two important properties of these matrices.